

Measuring Drift in Long Reading Sessions

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Abstract

We describe a small observational study of attention drift while reading technical papers on screen. Readers were asked to follow a single argument across several pages while we recorded where they paused, where they went back, and where they gave up. The results suggest that interruptions cluster around page turns and figure references rather than around difficult passages.

1 Introduction

Reading a research paper is rarely a linear activity. A reader moves back to an earlier definition, jumps ahead to a figure, and returns to the paragraph that cited it, often several times per page. Each of these moves carries a small cost, and the costs add up over a long session in ways that are easy to overlook when a tool is designed around single page views.¹

Earlier work on reading interfaces has mostly measured speed and recall on short passages. Those measures are useful, but they do not capture what happens in the fortieth minute of a dense proof, when the reader has already flipped between the same two pages a dozen times and the thread of the argument has started to fray.

1.1 Scope of this study

We restrict attention to papers typeset in a single column with numbered sections, because that layout makes page turns the dominant kind of interruption. Two-column layouts introduce a second kind of jump, from the bottom of one column to the top of the next, which we leave for a separate report.

¹We use the word session for one uninterrupted sitting with a single document.

2 Method

Each participant read one paper of about twenty pages while an observer logged every backward jump. We summarise a session by its drift score, which weights each jump by the distance travelled and discounts jumps that end on a figure or table:

$$D = \frac{1}{T} \sum_{k=1}^n w_k |p_k - p_{k-1}|. \quad (1)$$

Here the sum runs over all jumps in a session of length T minutes, and the weight is smaller for jumps that land on a float.

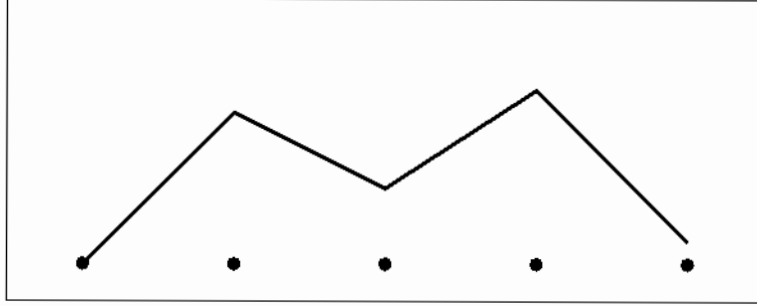


Figure 1: Drift score over time for one representative session. The score rises sharply after each page turn and decays slowly afterwards.

2.1 Participants and materials

Twelve graduate students took part. Each chose a paper from their own field that they had not read before, so that the material was neither trivial nor entirely unfamiliar. Sessions were held in a quiet office with the same monitor and the same reader software for everyone.